

Information and Estimation Theory
Lecture 05b: Channels

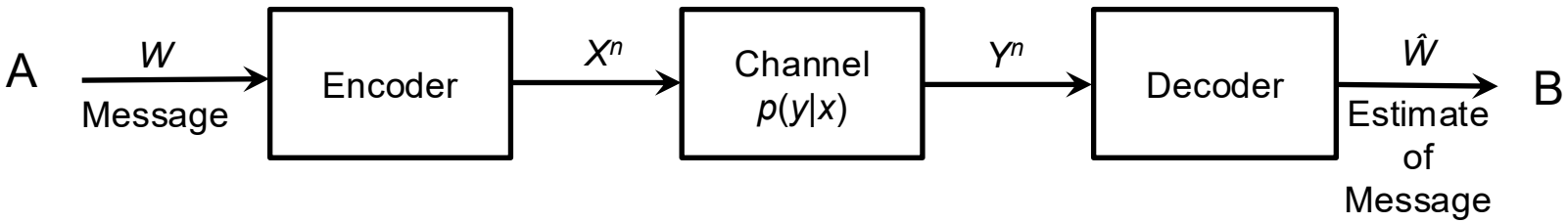
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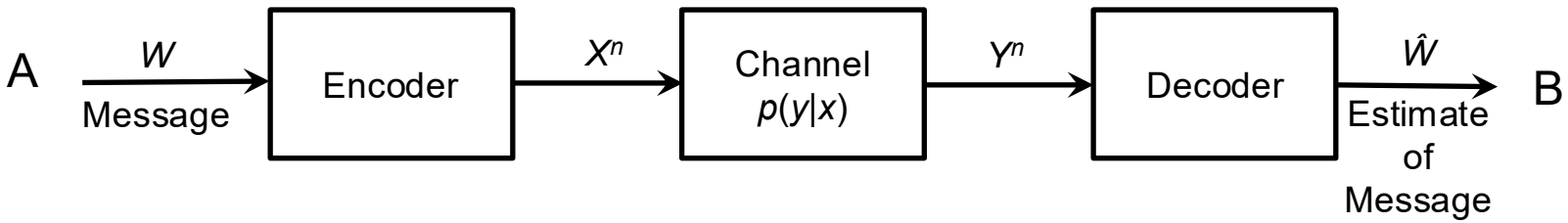
Communication Channels

- What do we mean when we say that A communicates with B?
  - The physical acts of A have induced a desired physical state in B
- Transfer of information is a physical process subject to uncontrollable ambient noise and imperfections
- If B and A agree on what was sent, then communication is successful



- Source symbols from a finite alphabet compose the message W
- W is mapped (*encoded*) into a sequence of n channel symbols X^n, which traverse the channel
- The channel introduces uncertainty due to communication imperfections, modeled by the conditional PMF $p(y|x)$

Channels and Confusion

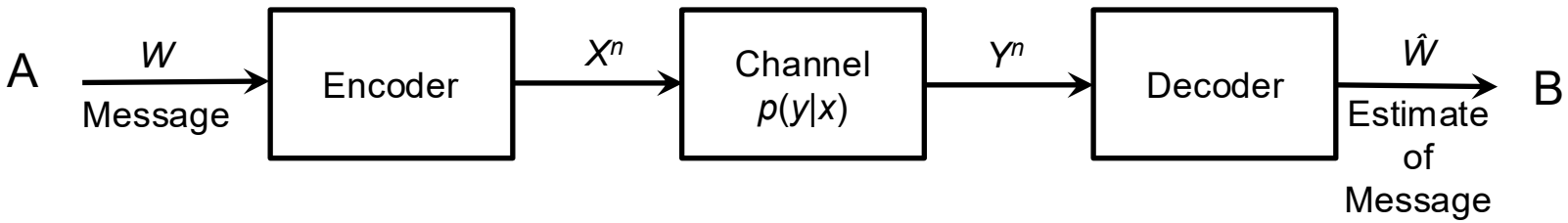


- Thus, the output sequence Y^n is further randomized, but has a distribution that depends on the input sequence
- From the output sequence Y^n, we attempt to recover (*decode*) the transmitted message W through its estimate $\hat{W}$
- Each of the possible input sequences induces a probability distribution on the output sequences, such as $p(\hat{W}|W)$
- Since two different input sequences may give rise to the same output sequence, the inputs are *confusable*

Message 1: «The cat sat on the mat»
Message 2: «The cat sat on the map»

Received: «The cat sat on the mad»
Estimate: «The cat sat on the ma?»

Channels and Capacity



- We can choose a *nonconfusable* subset of input sequences so that, with high probability, there is only one highly likely input that could have caused that output
  - We can reconstruct the input with a negligible probability of error

Message 1: «The cat sat on the mattt»
Message 2: «The cat sat on the mappp»

Received: «The cat sat on the madpp»
Estimate: «The cat sat on the map»

- By mapping the source into the appropriate "widely spaced" input sequences to the channel, we can transmit a message with very low probability of error and reconstruct it at the output
- The maximum rate at which this can be done is called the **capacity of the channel**

Capacity of a Discrete Memoryless Channel

- **Discrete channel:** system with input and output alphabets $\mathcal{X}$ and $\mathcal{Y}$ and a transition matrix $p(y|x)$ with the probabilities of observing the output symbol y given that we send the symbol x
- The channel is **memoryless** if the PMF of the output is conditionally independent of previous channel inputs or outputs

- The **information channel capacity *C*** of a discrete memoryless channel (DMC) is

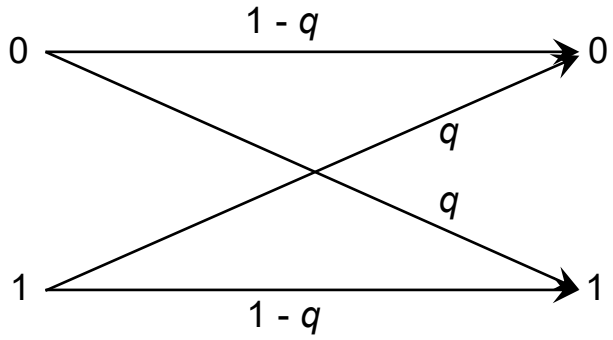
$$C = \max_{p(x)} I(X;Y)$$

  - The maximum is taken over all possible input PMFs $p(x)$

- The **operational channel capacity** is the highest rate at which information can be sent with arbitrarily low probability of error
  - Shannon → information channel capacity = operational channel capacity

Binary Symmetric Channel (BSC)

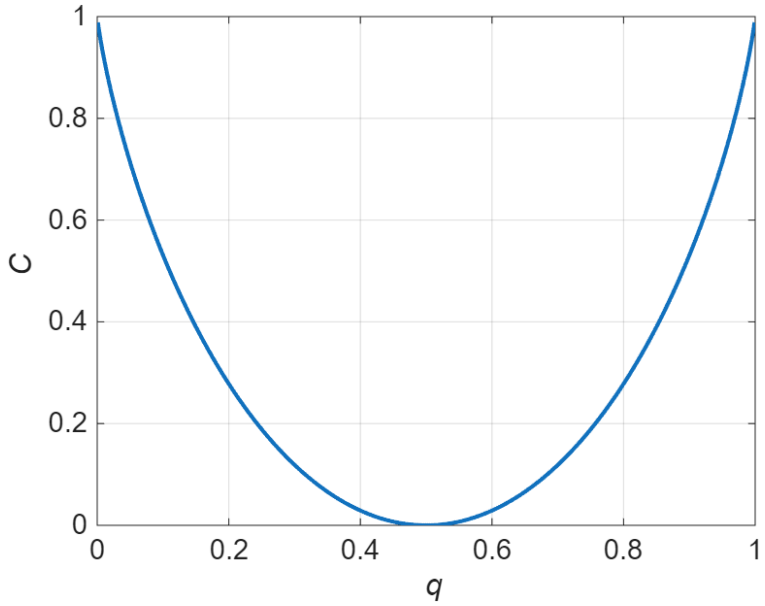
- Binary DMC in which the input symbols are flipped with probability q
  - This is the simplest model of a channel with errors
  - When an error occurs, a 0 is received as a 1, and vice versa



- The received bits do not reveal where the errors have occurred
  - All the received bits are unreliable

$$
\begin{aligned}
I(X;Y) &= H(Y) - H(Y|X) \\
&= H(Y) - \sum_x p(x) H(Y|X=x) \\
&= H(Y) - \sum_x p(x) H_{\mathrm{B}}(q) \\
&= H(Y) - H_{\mathrm{B}}(q)
\end{aligned}
$$

$$
C = \max_{p(x)} I(X;Y) = \max_{p(x)} H(Y) - H_{\mathrm{B}}(q)
$$
$$
= 1 - H_{\mathrm{B}}(q) \text{ bits}
$$



Compression-Transmission Duality

- There is a duality between the problems of data compression and data transmission

- During data compression, we remove all the redundancy in the data to form the most compressed version possible
- During data transmission, we add redundancy in a controlled fashion to combat errors in the channel

- It can be shown that a general communication system can be broken into two parts and that the problems of data compression and data transmission can be considered separately